

MLR Institute of Technology

Dundigal, Hyderabad

Department of Electronics and Communication Engineering

AI-Readable Knowledge Base Document

1. Department Overview

The Department of Electronics and Communication Engineering (ECE) was established in 2005 with an initial intake of 60 students. In 2015, the institution was granted Autonomous Status. The department is accredited by the National Board of Accreditation (NBA) and is affiliated with active professional bodies including IETE, ISTE, IEEE, and IUCEE.

The department is staffed by highly qualified and experienced teaching faculty and technical staff. It runs an Innovative Projects Cell to nurture creativity and innovation, as well as an Entrepreneurship Development Cell to guide students interested in starting ventures. Gap labs are held between 4 PM and 6 PM daily to encourage research activities.

2. Vision and Mission

Vision

To provide quality technical education with innovation and emphasis on Research and Development, thereby fulfilling the needs of society, and to achieve Academic Excellence in Electronics and Communication Engineering for globally competent engineers.

Mission

- M1: To adopt innovative student-centric learning methods.
 - M2: To develop an orientation towards a futuristic view through research.
 - M3: To enable students to compete at national and international levels.
 - M4: To strengthen core competencies among learners through an experiential curriculum.
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3. Programs Offered

Undergraduate Program

Degree: B.Tech in Electronics and Communication Engineering

Duration: 4 Years

Student Intake: 240 students per year

Postgraduate Programs

Program 1: M.Tech in Embedded Systems

Intake: 24 students

Program 2: M.Tech in Digital Systems and Computer Electronics (DSCE)

Intake: 24 students

4. Accreditation and Recognition

- Accreditation by National Board of Accreditation (NBA) for 3 years in 2013.
 - Re-accreditation by NBA for 3 years in 2016.
 - Autonomous Status granted to the institution in 2015.
 - EAPCET Code: MLID
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5. Program Outcomes (POs)

The following Program Outcomes are defined for the B.Tech ECE program:

- PO 1 - Engineering Knowledge: Apply knowledge of mathematics, science, engineering fundamentals, and specialization to solve complex engineering problems.
- PO 2 - Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems to reach substantiated conclusions.
- PO 3 - Design/Development of Solutions: Design solutions for complex engineering problems, considering public health, safety, and environmental factors.
- PO 4 - Conduct Investigations of Complex Problems: Use research-based knowledge including experiment design, data analysis, and synthesis to provide valid conclusions.
- PO 5 - Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools for complex engineering activities.
- PO 6 - The Engineer and Society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to engineering practice.
- PO 7 - Environment and Sustainability: Understand the impact of engineering solutions in societal and environmental contexts and demonstrate the need for sustainable development.
- PO 8 - Ethics: Apply ethical principles and commit to professional ethics and responsibilities.

- PO 9 - Individual and Team Work: Function effectively as an individual and as a member or leader in diverse and multidisciplinary teams.
 - PO 10 - Communication: Communicate effectively on complex engineering activities, including written reports, design documentation, and presentations.
 - PO 11 - Project Management and Finance: Demonstrate knowledge of engineering management principles and apply them in team and project settings.
 - PO 12 - Life-Long Learning: Recognize the need for and engage in independent and life-long learning in the context of technological change.
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6. Program Specific Outcomes (PSOs)

- PSO 1: Proficiency in specialized software tools and computer programming for the analysis and design of Embedded Systems and VLSI.
 - PSO 2: Capability to comprehend technological advancements in Radio Frequency (RF), Communication, and Digital Signal Processing.
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7. Research Areas

The department has a number of active PhD scholars working in the following research areas:

- Image and Video Processing
 - Signal Processing
 - Embedded Systems
 - Nano Materials
 - Microwave Communications
 - VLSI Technology
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8. Laboratories and Equipment

Key Laboratory Equipment

- PCB Design Machine
- Spectrum Analyzer
- Digital Storage Oscilloscopes
- Spartan 3E (FPGA) Kits
- TMS 320C6748 DSP Kits with Code Composer Studio

Laboratory Software

- MULTISIM
- VHDL
- Xilinx ISE Foundation
- Cadence
- MATLAB with Toolboxes

9. Centres of Excellence (COEs)

The department has four Centres of Excellence. Their objective is to facilitate and promote Research and Development.

COE 1: Embedded and Robotics

Focuses on embedded systems, robotics, FPGA-based projects, and IoT applications. Conducts Faculty Development Programs and workshops on Real-Time Embedded Systems and IoT.

COE 2: Cadence

Focuses on VLSI design and analog/digital IC design. Objective: To train students in VLSI tools and prepare them for careers in VLSI companies. Conducts workshops on Custom IC design flow, ASIC design flow, and Cadence Tools.

COE 3: MATLAB

Focuses on hardware connectivity with MATLAB and simulation. Conducts Faculty Development Programs and workshops on hardware implementation with MATLAB.

COE 4: LabVIEW Academy

Focuses on Virtual Instrumentation using LabVIEW. Conducts training on LabVIEW and its integration with hardware platforms including Arduino and Raspberry Pi.

10. Faculty Development Activities

The department regularly organizes and participates in Faculty Development Programs (FDPs) and workshops. Notable programs include:

MATLAB-Related Programs

- 3-Day FDP on Hardware Connectivity with MATLAB and Simulation (May 27-29, 2019). Resource persons: Mr. Nitin Rai and Mr. Prem Kumar J from MATHWORKS India Pvt Ltd, along with Dr. B. Sridhar and Dr. T. S. Arulananth from MLRIT. Attended by 20 faculty from MLRIT, 5 from MLRITM, and 3 from BVRIT.
- 1-Day Workshop on Hardware Implementation with MATLAB (December 16, 2017). Resource person: J. Prem Kumar, Product Manager, Capricot Technologies Pvt. Ltd., Hyderabad.
- 1-Day FDP on MATLAB Interfacing with Arduino and Raspberry Pi (November 23, 2016).

Cadence-Related Programs

- 2-Day Workshop on Cadence Tools (December 11-12, 2017). Sessions on Custom IC Design Flow and ASIC Design Flow. Trainer: Mr. Sharth Kanth, Senior Support Engineer, ENTUPLE Technology, Bangalore.

- 2-Day FDP on Analog and Digital IC Design using Cadence (March 8-9, 2017). Trainers from Entuple Technologies, Bangalore, and Dr. P. Chandrashekar, Professor, MLRIT.

Embedded Systems and Robotics Programs

- 4-Day FDP on Real-Time Embedded Systems (FPGA Boards) and IoT Applications (December 5-8, 2017).
- Workshop on Embedded System Design using AVR Microcontrollers (April 17-22, 2017). Trainer: Mr. Sarath Kumar, Robo Labs Pvt. Ltd., Pune.

Signal Processing and Communications Programs

- 3-Day Workshop on Digital Signal Processors (May 5-8, 2017). 27 faculty attended from MLRITM, IARE, and MLRIT.
- 3-Day Workshop on Analog and Digital Communications (May 15-17, 2017). 20 faculty from MLRIT and MLRITM. Trainer: Dr. B. Sridhar, Professor, ECE, MLRIT.
- 3-Day Workshop on Microprocessors and Microcontrollers (May 3-6, 2017). Trainer: Mr. Kiran Kumar, Co-founder, Professional Advanced Electronics, Hyderabad.

PCB Design Programs

- 3-Day Workshop on PCB Design (May 11-13, 2017). 16 faculty from MLRIT and 7 from MLRITM trained by Dr. Mahendra V, Mr. Naga Raju, Mr. E. Amareswar, and Mr. Raju Naik, all from ECE, MLRIT.

Other Faculty Programs

- 3-Day Workshop on Digital Image Processing using NI Vision System (June 28-30, 2017). Trainers: Mr. Vasanth Kumar V and Mr. Suresh, Senior Application Engineers, VI Solutions, Bangalore.
- FDP on Instructional Design and Delivery System by NITTTR (July 2-7, 2018).
- 2-Day FDP on Virtual Instrumentation using LabVIEW (October 21-22, 2016).
- Guest Lecture on Controls, Robotics and Mechatronics (April 17, 2017). Speakers: Mr. Yona Baskharoun, Regional Manager, Quanser; Mr. Visweswaran Jagadeesan, NI Bangalore.
- Workshop at National Remote Sensing Centre, Hyderabad on Science Day (February 28, 2017).

11. Student Activities

Micro Project Competitions

Interdepartmental Micro Project Competitions were organized regularly under the IETE Student Forum (ISF) and Center for Innovation and Entrepreneurship (CIE). Notable competitions:

- Competition on April 27, 2019, evaluated by Dr. M. Srinivasa Rao (BITS-Pilani) and Dr. Narasimham (Senior Engineer, BARC, Hyderabad). First Prize: Shop Online Trail Online (Mentor: Mr. Ramesh). Second Prize: Home Automation (Mentor: Mr. Laxma Reddy).
- Competition on November 23, 2019. First Prize: Smart Energy Meter (Mentor: Mr. A. Laxmana Siridhar). First Prize: Enhancement for Village Income (Mentor: Mr. K. Nishanth Rao).

- Micro Project Competition held on November 2 and 10, 2016 for ECE students.

Innovative Ideas Competition

MLR Institute of Technology conducted an Innovative Ideas Competition on September 21, 2019, themed Farmers and Latest Technologies. Twenty batches participated. Awards were presented by Dr. S. V. S. Prasad (HOD ECE) and Dr. V. Mahendra (Professor ECE).

Student Workshops

- Workshop on PCB Designing for II year ECE students (August 29, 2017) under ISTE Student Chapter.
- Workshop on Satellite Image Processing using ENVI Software (July 2, 2016) by Mrs. V. Satya Sahithi, ISRO Hyderabad.
- Skill Development Workshop on PCB Fabrication (February 8, 2017) for Diploma students under IETE Student Forum.

Guest Lectures for Students

- Guest Lecture on Electronic Devices and Circuits for II year ECE by Dr. K.V.N.S.V.P.L. Narasimham, Scientist BARC and Chairman, Technical Program Sub-Committee IETE (July 29, 2017).
- One-Day Guest Lecture on Electronic Devices and Circuits (August 25, 2018). Delivered by Dr. K. V. N. S. V. P. L. Narasimham, Scientific Officer (F), BARC, to 190 students.
- Guest Lecture on Satellite Communication for IV B.Tech students (January 17, 2017) by Mr. Terala Lingam, Scientist, ISRO.
- Dr. C.V.S. RAO 4th Memorial Lecture on High Power Radar Studies of Atmosphere and Ionosphere (August 22, 2017) at IETE Hyderabad Centre. Speaker: Dr. Pendyala Balarama Rao, Former Director NARL.

Induction and Orientation Programs

Induction Program conducted by the Department of ECE on July 28, 2018, for first-year students of ECE and EEE departments.

Industrial Visits

Forty students visited the INFOSYS Gachibowli campus on January 9-10, 2017. The visit was led by Mr. M. Raju Naik and Mr. V. Sai Kumar Reddy, Assistant Professors, ECE, MLRIT.

NI-CLAD Certification Achievements

- 21 students of 2nd year ECE passed the NI-CLAD exam on March 18, 2017.
- 15 students of 3rd year ECE passed the NI-CLAD exam on March 18, 2017.
- 29 students of 4th year ECE passed the NI-CLAD exam on March 3, 2017.

Engineer's Day Celebrations

Engineer's Day was celebrated on September 14 and 16, 2016. Events included Poster Presentation, Quiz, Paper Presentation, Robotic Workshop, Crossword, and Circuit Debugging.

12. Student Achievements and Awards

Major Competition Wins

- 1st Prize (Rs. 30,000) at RESEARCH EXPO, SHASTRA-2018, IIT Madras.
- 1st Prize (Rs. 30,000) at ANVESHANA-2017, Hyderabad (5th Edition).
- 1st Prize at EXHIPRO, SPOORTHY 2018, JNTUH, Hyderabad.
- 1st Prize at E-PAPYRUS, SPOORTHY 2018, JNTU-CEH.
- 1st Prize at HAVANA, National Level Intercollegiate Technical Championship, GITAM University, Hyderabad.

IEEE Expo Achievements (2017)

- 1st Prize: B. Sai Prakash and M. Lavanya Sri (3rd year ECE-B) for project Quad Wheel Omni Directional Robot at IEEE Engineering College Expo, Nalla Narsimha Reddy Group of Institutions (February 18, 2017). Guided by Mr. E. Amareshwar and Mr. J. Chandra Shekhar, ECE MLRIT.
- 2nd Prize: Ms. N. Sree Nanda, Mr. N. Bineeth Patil, Ms. Sai Lakshmi, and Mr. Vishnu Kumar (4th year ECE-C) for project Child Rescue and Pipeline Inspection Robot at IEEE Engineering College Expo (February 18, 2017). Guided by Mr. Raju Naik and Ms. Jyothi K, ECE MLRIT.

Other Student Awards

- Consolation Prize: Uma Surya Sai and Samyuktha (3rd year ECE-A) for project Digital Watermarking at ANVESHANA Science and Engineering Fair, January 24, 2017. Guided by Dr. A. V. Paramkusam, HOD ECE.
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13. Publications and Research Output

- Faculty and students have published over 80 research papers in Scopus-indexed international journals.
- Six research papers were accepted and presented at the World Summit on Science, Engineering and Technology (Cambridge Summit 2018), held at the University of Cambridge, UK (January 4-6, 2018).

Faculty Recognition

- Dr. Karthik Rajendra, Professor of ECE, was felicitated as a Session Chair at the Cambridge Summit 2018.
 - Dr. Arulananth, Professor of ECE, received the Bharat Vikas Award from the Institute of Self Reliance on November 19, 2017, for loyalty, diligence, and outstanding performance in Communication Engineering.
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14. Roll of Honour

B.Tech Roll of Honour

S.No.	Year	Hall Ticket Number	Name of Student	Percentage / CGPA
1	2009	05R21A0416	B Mahesh Kumar	76%
2	2010	06R21A0455	P Uma Sadevi	83%
3	2011	07R21A0454	K Srutha Keerthi	85%
4	2012	08R21A0473	K Krishna Padma Sree	84%
5	2013	09R21A0437	Ramya Swetha	86%
6	2014	10R21A0479	G Sneha Latha	84%
7	2015	11R21A0486	G Roja Rani	80%
8	2016	12R21A04E2	K Lakshmi	85%
9	2017	13R21A04F5	Kiran Kumari	81%
10	2018	14R21A04A8	Reddy Sai Saranya	80%
11	2019	15R21A04F5	Nadimpally Lakshmi Sowmya	9.75
12	2020	16R21A0433	Kuntamukkala VSSS Vyshnavi	9.78
13	2021	17R21A04P0	Aka Divya Sree	9.45
14	2022	18R21A0454	Vankuru Charitha	9.49
15	2023	19R21A0446	Rongala Rohi	9.37
16	2024	20R21A04N6	Revurai Saiteja Reddy	9.43
17	2025	21R21A04G1	M Akhil Verdhan Reddy	9.13

M.Tech Roll of Honour

S.No.	Year	Specialization	Hall Ticket Number	Name of Student	CGPA
1	2017	DSCE	15R21D0607	K Nandini	9.05
2	2017	ES	15R21D5508	Malyala Pravallika	9.49
3	2017	ES	15R21D5515	Vidya Turlapati	9.49
4	2018	DSCE	16R21D0609	Satlapally Divya	9.00
5	2018	ES	16R21D5514	Peddapuram Vigneshwar Reddy	9.14
6	2019	ES	17R21D5514	Rahul S R	8.33
7	2020	DSCE	18R21D0603	Shaik Nasir Hussain	8.30
8	2020	ES	18R21D5505	P Raja Rajeshwari	8.40

9	2021	DSCE	19R21D0605	Nadimpally Lakshmi Sowmya	8.98
10	2021	ES	19R21D5515	Velpuri Voorwashi	8.91

15. Professional Bodies and Memberships

The department is an active member of the following professional bodies:

- IETE - Institute of Electronics and Telecommunication Engineers
- ISTE - Indian Society for Technical Education
- IEEE - Institute of Electrical and Electronics Engineers
- IUCEE - Indo-US Collaboration for Engineering Education

The department runs active student forums under IETE and ISTE student chapters, which regularly organize workshops, competitions, and guest lectures.

16. Sample Student Projects

Class	Student Names	Mentor	Project Title
II Yr ECE-C	J. Raj Kumar, N. Saikiran, T. Tejas	Mr. B. Annapurna	Smart Helmet
II Yr ECE-D	A. Chaitanya Sree, V. Amulya, C. Bhavana	Mr. Vasudeva Reddy Nandigama	Ultra Sonic Sight
II Yr ECE-C	B. Supriya, M. Nitishsai, Md. Karishma	Mr. Raju Naik	Obstacle Avoiding Robotic Vehicle
III Yr ECE-C	Harshitha N N, Akhila, P.E. Austin Paul	Mr. A. Laxmana Siridhar	Smart Energy Meter
III Yr ECE-D	R. Divya Sree, Teda Santhosh Kumar, P. Sai Teja	Mr. P. B. Natarajan	Smart Garbage Segregator
III Yr ECE-C	K. Deekshith Reddy, G. Likitha, K. Srinivas, Ch. Phani Sai	Mr. K. Nishanth Rao	Enhancement for Village Income

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